Momentum in Optimization

1 Normal Gradient Descent

Without momentum, the parameters are updated as

$$\boxed{\theta_{t+1} = \theta_t - \eta g_t}$$

where

$$g_t = \nabla_\theta \mathcal{L}(\theta_t).$$

- θ_t represents the model parameters at step t,
- η is the learning rate,
- g_t is the gradient of the loss function $\mathcal{L}$.

Standard gradient descent depends only on the current gradient.

2 Gradient Descent with Momentum

With momentum, we introduce a velocity term v_t:

$$\boxed{v_t = \beta v_{t-1} + g_t}$$

$$\boxed{\theta_{t+1} = \theta_t - \eta v_t}$$

where:

- g_t is the current gradient,
- v_t is the accumulated direction (velocity),
- β is the momentum coefficient, commonly around 0.9,
- η is the learning rate.

3 What Does Momentum Remember?

Expanding the velocity recursively gives

$$\begin{aligned} v_t &= \beta v_{t-1} + g_t \\ &= g_t + \beta g_{t-1} + \beta^2 g_{t-2} + \beta^3 g_{t-3} + \cdots . \end{aligned}$$

Therefore, the parameter update can be written as

$$\boxed{\theta_{t+1} = \theta_t - \eta \left(g_t + \beta g_{t-1} + \beta^2 g_{t-2} + \cdots \right)}$$

This shows the main difference between standard gradient descent and momentum:

Standard GD

Current gradient only

$$\theta_{t+1} = \theta_t - \eta g_t$$

Momentum

Current gradient + gradient history

$$\theta_{t+1} = \theta_t - \eta \left(g_t + \beta g_{t-1} + \beta^2 g_{t-2} + \cdots \right)$$

4 Role of Momentum

In optimization, **momentum remembers the direction of previous gradients** and combines them with the current gradient. Its main roles are:

Why it helps

- **Speeds up optimization** when gradients consistently point in the same direction.
- **Reduces oscillation** when gradients repeatedly change direction.
- **Provides smoother optimization** through noisy or relatively flat regions.

5 Key Intuition

Momentum behaves somewhat like a ball rolling downhill. A ball does not determine its movement only from the slope at its current position. It also carries velocity accumulated from its previous movement.